

VISHESH KUMAR

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SUMMARY

PhD Scholar in Data Science and Engineering at IISER Bhopal with research experience in computer vision, adversarial robustness, explainable AI, secure and trustworthy AI, and foundation models. Published in CVPR, ECCV, AISTATS, IEEE Transactions on Big Data, IJCB, and AAAI Student Abstract track. Interested in research and applied machine learning roles focused on robust vision systems, multimodal learning, and secure AI. Currently developing an agentic AI-based robustness auditing tool that automatically evaluates vision models against adversarial attacks and generates structured vulnerability reports.

EDUCATION

Indian Institute of Science Education and Research (IISER) Bhopal *Dec 2022 – Present*
PhD, Data Science and Engineering

Indian Institute of Engineering Science and Technology (IIST) Shibpur *Jul 2019 – Aug 2021*
M.Sc. in Applied Mathematics *CGPA: 8.8/10*

Motilal Nehru College, University of Delhi *Jul 2016 – Jun 2019*
B.Sc. in Physical Science *CGPA: 7.71/10*

RESEARCH INTERESTS

- Computer Vision, Explainable AI, Secure and Trustworthy AI
- Adversarial Robustness, Deep Learning, Machine Learning
- Foundation Models, Deepfakes, Presentation Attack Detection
- Diffusion Models
- Agentic AI

RESEARCH EXPERIENCE

Visiting Scholar, Rice University, Houston, Texas, USA *Jan 2026 – June 2026*

- Working under the supervision of Dr. Guha Balakrishnan.
- Developing an interpretable algorithm to localize the adversarial patches and secure the vision system to make a trustworthy AI.

PhD Scholar, Department of Data Science and Engineering, IISER Bhopal *Dec 2022 – Present*

- Conduct research in trustworthy AI, adversarial robustness, explainable AI, and computer vision.
- Develop methods for adversarial patch detection, robustness benchmarking, and secure model evaluation.
- Publish work at venues including CVPR, AISTATS, IEEE Transactions, IJCB, and AAAI.

Junior Research Fellow, DA-IICT, Gandhinagar *Aug 2022 – Dec 2022*

- Worked on the SERB-funded project “Dynamical Sampling and Representation of Frames via Iterated Operator System” under the guidance of Prof. Nabin Kumar Sahu.

Master’s Thesis, IIST Shibpur

- Banach's Contraction Principle and its Extensions

SELECTED PUBLICATIONS

1. V. Kumar, G. Balakrishnan, and A. Agarwal, *Breaking Rigidity in Adversarial Patch Attacks*, 2025. European conference on computer vision (ECCV), 2026.
2. V. Kumar, A. Agarwal, *Feature Collapse Under Corruption: An Entropy Perspective on Robust Neural Networks*, International Conference on Machine Learning (ICML), 2026.
3. V. Kumar, A. Agarwal, *The Unseen Adversaries: Robust and Generalized Defense Against Adversarial Patches*, International Conference on Artificial Intelligence and Statistics (AISTATS), 2026.
4. V. Kumar, S. Shukla, A. Agarwal, *Robustness Benchmarking of Convolutional and Transformer Architectures for Image Classification*, IEEE Transactions on Big Data, 2025. [Link](#)
5. V. Kumar, A. Agarwal, *A Unified, Resilient, and Explainable Adversarial Patch Detector*, IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025. [Link](#)
6. V. Kumar, A. Chanda, P. Bhattacharya, A. Agarwal, *Q-MoFusion: A Quantum Classifier for Mosquito Species Classification*, AAAI Conference on Artificial Intelligence, Student Abstract Track, 2026.
7. V. Kumar, A. Agarwal, *Are Object Recognition Models Effective and Unbiased for Biometric Recognition?*, IEEE International Joint Conference on Biometrics (IJCB), 2024. [Link](#)
8. V. Kumar, A. Agarwal, *On Unconstrained Ear Recognition for Privacy-Preserving Authentication*, Workshop on Advances of Mobile and Wearable Biometrics (WAMWB), in conjunction with ACM MobileHCI, 2023. [Link](#)

MANUSCRIPTS UNDER REVIEW & IN SUBMISSION

1. V. Kumar, G. Balakrishnan, and A. Agarwal, *Interpretable by Construction: Unified Detection and Localization of Adversarial Patches*, 2026. Under Review.
2. N. K. Sahu, V. Rajput, *Dynamical Representation of Frames in Tensor Product of Hardy Spaces*, arXiv:2308.11330, 2023. Under minor revision in La Matematica. [Link](#)

AWARDS AND FELLOWSHIPS

- ICML Travel Award for ICML 2026
- ACM/IARCS Travel Grant for AISTATS 2026
- Mehta Rice Engineering Scholars Program (MRESP), Rice University, 2025–2026
- CVPR 2025 DEI Travel Grant
- ACM/IARCS Travel Grant for CVPR 2025
- ConfAI 2025 Travel Grant
- Visvesvaraya PhD Fellowship, Government of India
- Qualified IIT JAM (2019, 2020) and GATE (2022)

CURRENT PROJECTS

Agentic AI Tool for Automated Robustness Evaluation of Vision Models

2026 – Present

- Developing an agentic AI-based evaluation platform where users provide a dataset and model checkpoint, and the system automatically selects and runs adversarial attacks to identify failure cases.
- Designing an automated robustness pipeline that evaluates model performance under adversarial perturbations, patch attacks, corruptions, and other input-level stress tests.
- Building an end-to-end reporting module that summarizes attack success rates, accuracy degradation, failure examples, and model vulnerabilities in a structured technical report.
- Aiming to support researchers and practitioners in auditing deployed vision models for security, robustness, and trustworthy AI.